

FULL COURSE

Senior Systems Engineer

Distributed systems, SRE, reliability engineering and architecture at scale

17 6–8 months

Advanced

Live+Recorded

Prasant Mishra

ABOUT THIS COURSE

The most advanced systems program at Qurious Academy. Covers distributed systems architecture, SRE fundamentals, reliability engineering, chaos engineering, and production operations at scale. Designed for engineers targeting Staff/Principal-level systems roles.

Prerequisites: Systems Engineer or equivalent 3+ years in infrastructure/platform engineering

WHAT YOU'LL LEARN

- 1 Design and operate high-availability distributed systems
- 2 Lead SRE practices across engineering teams
- 3 Run chaos experiments and improve system resilience
- 4 Architect production systems at scale
- 5 Succeed in Staff/Principal systems interviews

CURRICULUM · 6 SESSIONS

SESSION MODULE 1 — MONTHS 1–2

Distributed Systems

CAP theorem, consistency models

Distributed consensus — Raft & Paxos

Distributed transactions & SAGA pattern

Event sourcing & CQRS

SESSION MODULE 2 — MONTHS 3–4

High-Availability Architecture

Multi-region deployments

Blue-green & canary deployments

Database replication & failover

Load balancing at scale

SESSION MODULE 3 — MONTHS 5–6

SRE Fundamentals

SLOs, SLAs, error budgets

Incident management & post-mortems

Toil elimination & automation

On-call strategy & runbooks

SESSION MODULE 4 — MONTHS 7–8

Reliability Engineering

Chaos engineering — principles & practice

Game days & failure injection

Resilience patterns — circuit breakers, retries, bulkheads

Performance engineering & profiling at scale

SESSION MODULE 5 — MONTHS 9–10

Observability at Scale

Distributed tracing — OpenTelemetry, Jaeger

Metrics at scale — Prometheus, VictoriaMetrics

Log aggregation — Loki, ELK stack

Capacity planning & forecasting

SESSION MODULE 6 — MONTHS 11–12

Architecture & Leadership

Architectural decision records (ADRs)

Technical roadmapping

Staff-level interview prep

Capstone: design a production system from scratch

WHAT'S INCLUDED

✓ Weekly live sessions + all recordings

✓ Chaos experiment playbooks

✓ 1-on-1 mentorship sessions

✓ Completion Certificate

✓ Architecture case studies

✓ Staff-level mock interviews

✓ Alumni network access

UPCOMING BATCHES



Ongoing · Sat & Sun · 10:00 AM – 1:00 PM IST



New batches start monthly



completion

₹1,19,999

Live+Recorded · 6–8 months

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